

General Knowledge

Indian Geography

STUDY NOTES

Soils in India



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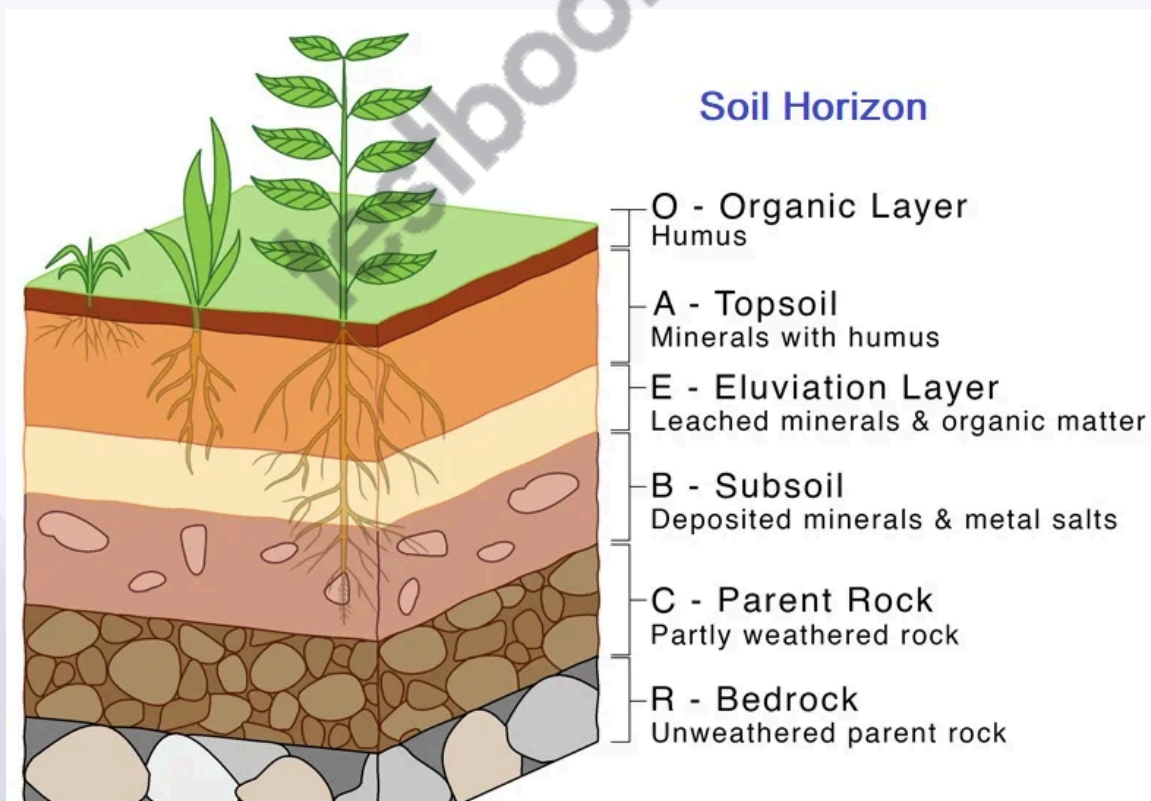
Soil is the thin layer at the top of the Earth's crust, formed by the breakdown of rocks due to wind, water, and glacial actions. It contains rock particles enriched with organic content, providing a base for life and agriculture. The scientific study of soils in their natural environment is called **Pedology**. The process by which soil is formed through weathering, leaching, and calcification is known as **Pedogenesis**.

Factors Affecting Soil Formation

1. **Climatic Conditions:** Temperature and rainfall patterns shape soil types.
2. **Surface Relief:** Topography affects soil erosion, composition, and drainage.
3. **Natural Vegetation:** Influences the soil's organic content and fertility.
4. **Rock Material:** The parent rock determines the minerals, color, and texture of the soil.

Soil Profile

The **soil profile** is a vertical cross-section of soil layers, showing variations in color, depth, chemical composition, and texture. Each layer is known as a **Soil Horizon** and has unique characteristics.



Types of Soils in India

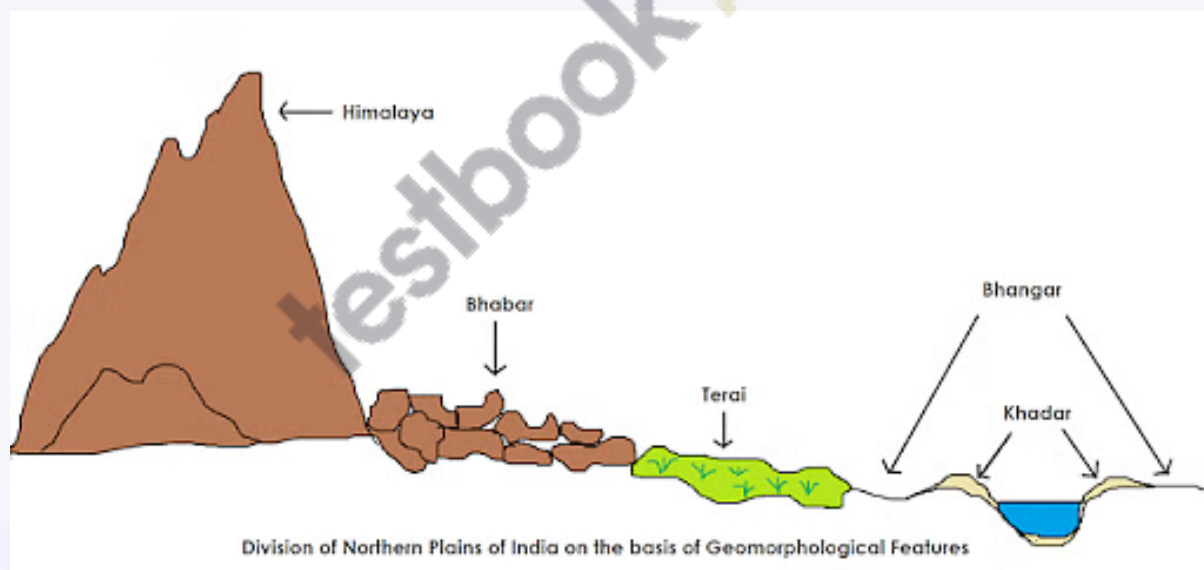
According to a substantial study and research by ICAR (Indian Council of Agricultural Research), soils in **India have been classified into eight different categories**, which are :

- Alluvial soil
- Black soil
- Red soil
- Laterite soil
- Forest and mountain soil
- Desert soil
- Saline or alkaline soil
- Marshy soil or bog soil



Alluvial soil

- The majority of the accessible soil in India, accounts for around **43 percent** and spans **143 square kilometers**.
- Common throughout the **river basins** and plains of the **northern regions**.
- **Deltas** and **estuaries** are where you are most likely to find them in **peninsular India**.
- There is a presence of **humus, lime, and organic materials**.
- Extremely **productive**.
- Examples include the **Indus-Ganga-Brahmaputra plain, Narmada-Tapi plain**, etc.
- They are a type of soil that forms as a result of rivers, streams, and other waterways transporting and depositing material.
- The **western** part of the country has a higher proportion of **sand** than the eastern part.
- **Old alluvium** is referred to as **Bhangar**, whereas more **recent alluvium** is referred to as **Khadar**.
- Shades of **light grey** all the way to **ash grey**.
- **Sandy to silty loam or clay** is the predominant texture.
- **Potash-containing**; plentiful.
- Having a lack of **phosphorus**.
- The primary crops grown include **wheat, rice, maize, sugarcane, legumes, and oilseeds**, among others.



Black/Regur soil

- The word "**regur**" refers to **cotton** and is the ideal soil for growing **cotton**.
- **Black soil** makes up the majority of the **Deccan's** landscape.
- **Mature soil**.

- A high capacity for holding onto **water**.
- When wet, it will **swell** and get sticky, but when it's dry, it will **shrink**.
- The black soil is characterized by the fact that it is capable of **self-ploughing** since, when dry, it develops broad **fissures**.
- Rich in: **Iron, lime, calcium, potassium, aluminium, and magnesium**.
- **Nitrogen, phosphorus, and organic materials** are all absent from the sample.
- The colour ranges from a very **dark black** to a **lighter black**.
- **Clay-like** in its consistency.

Red soil

- Mostly observed in areas with **minimal annual precipitation**.
- Alternately referred to as the **Omnibus group**.
- **Porous, friable** structure.
- **Kankar** and **lime** are not present (impure calcium carbonate).
- **Lime, phosphate, manganese, nitrogen, humus, and potash** are not present in sufficient amounts.
- As a result of **ferric oxide**, the colour is **red**. The lower layer is either **yellow** or a **yellowish-reddish** colour.
- **Sandier** to **clayier** and more **loamy** in texture.
- It is possible to cultivate things like **wheat, cotton, legumes, tobacco, oilseeds, potatoes**, and so on.

Laterite soil

- The name is derived from the **Latin word** for brick, which is "**later**".
- Turns into something that is extremely **soft** when wet but extremely **hard** when dry.
- Found in regions characterised by both **high temperatures** and **heavy precipitation**.
- Resulting from a significant amount of **leaching**.
- **Lime** and **silica** will be removed from the soil as a result of leaching.
- Because of the high temperature, the **bacteria** in the soil will swiftly extract any organic matter that is present, and the **humus** will be quickly consumed by the trees and other plants. As a result, the amount of **humus** is low.
- **Iron** and **aluminium** are found in abundance.
- Shortcomings in the following nutrients: **nitrogen, potash, potassium, lime, and humus**.
- Coloring: **Red** colour owing to **iron oxide**.
- The majority of the land is used for cultivating **rice, ragi, sugarcane, and cashew nuts**.

Forest and Mountain soil

- **Heterogeneous soils** on **forest-covered hilly slopes** wrap **8.7%** of the whole area of the country's land.
- These are found generally in the **hilly** and **mountainous regions** where there is sufficient availability of **rainforests** on the slopes of the **eastern** and **western ghats**.
- They are rich in **humus**.
- It has a **slitty** and **loamy** texture, and this soil varies in different regions depending on the **climatic conditions** and **organic matter deposition**.
- It is suitable for cultivating **tea**, **coffee**, and **tropical fruits**.

Desert soil

- It is mainly deposited with the help of **wind action** and is found in the **Aravalli region** of **Rajasthan**, **northern Gujarat**, and some regions of **Saurashtra**.
- There is a high concentration of **soluble salts** but a comparatively low **moisture content**.
- These are generally rich in various **minerals** such as **phosphate** and **calcium**.
- These kinds of soils are suitable for the cultivation of **barley**, **maize**, **millet**, and **bajra**.

Saline or alkaline soil

- The soil at the top of **alkaline soil** is saturated with **alkaline** and **saline efflorescence** as a result of which it gets roofed with particles of **salt**.
- The soil enters the layer of **subsoil** in areas having low **water tables**. **Salts** get carried away by river water in areas having a good **drainage system**.
- These are also termed **Usara soil**. It is generally found in **western Gujarat** and the **Sunderban area** of **West Bengal**.
- It is **mineral-rich** with a large concentration of **potassium**, **sodium**, and **magnesium**, and thus, they are termed **infertile**.
- It consists of a **sandy** and **loamy** texture.
- They lack **nitrogen** content, and **seawater incursion** is the reason for the occurrence of **alkaline soil**.

Marshy soil or bog soil

- It is found near the **delta regions** and in some districts of **Kerala** and **Uttarakhand**.
- It is sufficiently rich in **organic material** and has a high **salinity quotient**.
- **Clay** and **mud** make this soil **heavy**, and it is also rich in **moisture content**.
- It is suitable for the cultivation of **jute** and **rice**.